

AI-Powered Customer Support Agent using n8n, Amazon Bedrock, Gmail, and Google Sheets

Overview

This solution automates customer support using AI.

The workflow receives customer emails, analyzes them using Amazon Bedrock, retrieves company-specific information from a Google Sheets knowledge base, generates a professional response, and optionally sends the response back to the customer.

Architecture

Customer Email ↓ Gmail Trigger ↓ Edit Fields ↓ AI Agent ↓ Amazon Bedrock Claude ↓ Google Sheets Tool ↓ Gmail Send Message

Components Used

- Gmail Trigger
 - Edit Fields
 - AI Agent
 - Amazon Bedrock Chat Model (Claude)
 - Google Sheets Tool
 - Gmail Send Message
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Step 1: Gmail Trigger

Purpose

Monitor an inbox for incoming customer emails.

Configuration

- Authentication: Gmail Account
- Event: New Email
- Simplify Output: Enabled

Example Email

Subject: Refund Request

Body: When can I get my refund?

Step 2: Edit Fields

Purpose

Prepare structured input for the AI Agent.

Fields Created

sender_email

Extract sender email address.

Expression:

```
{{json["From"].match(/<(.*)>/)[1]}}
```

email_subject

```
{{json["Subject"]}}
```

email_body

```
{{json.snippet}}
```

Output

```
{
  "sender_email": "customer@example.com",
  "email_subject": "Refund Request",
  "email_body": "When can I get my refund?"
}
```

Step 3: AI Agent

Purpose

Act as the intelligent decision maker.

Prompt

You are a professional customer support assistant.

Read the customer's email and generate a professional response.

Use company policies from the knowledge base whenever available.

Customer Email: {{\$json.email_body}}

Step 4: Amazon Bedrock

Model

Claude 3 Sonnet

Purpose

- Understand customer intent
 - Generate responses
 - Use external tools
 - Follow business rules
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Step 5: Google Sheets Tool

Purpose

Provide company-specific business knowledge.

Example Sheet

Sheet Name:

Company_Knowledge_Base

Question	Answer
Refund Policy	Refunds allowed within 30 days
Shipping Time	Delivery within 5 business days
Cancellation Policy	Orders can be cancelled before shipment
Contact Support	support@company.com

Operation

Get Row(s)

Step 6: AI Reasoning

Customer Email:

"When can I get my refund?"

Agent Actions:

1. Understand intent.
2. Identify refund-related query.
3. Retrieve refund policy from Google Sheets.
4. Generate response.

Example Response:

Dear Customer,

Thank you for contacting us.

Refunds are available within 30 days of purchase. Please share your order number for further assistance.

Best Regards, Customer Support Team

Step 7: Gmail Send Message

Purpose

Automatically send the generated response.

To

```
{{node["Edit Fields"].json.sender_email}}
```

Subject

```
Re: {{node["Edit Fields"].json.email_subject}}
```

Message

```
{{node["AI Agent"].json.output}}
```

End-to-End Flow

1. Customer sends an email.
 2. Gmail Trigger detects the email.
 3. Edit Fields extracts required information.
 4. AI Agent receives the request.
 5. Amazon Bedrock Claude analyzes the email.
 6. Google Sheets provides company knowledge.
 7. Claude generates a response.
 8. Gmail sends the response.
 9. Customer receives an AI-generated reply.
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Business Benefits

- Faster customer response times
 - Reduced manual effort
 - Consistent support responses
 - 24/7 availability
 - Easy integration with existing systems
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AWS Concepts Demonstrated

- Amazon Bedrock
 - Foundation Models
 - Agentic AI
 - Tool Use
 - Retrieval from External Knowledge Sources
 - Workflow Automation
 - Event-Driven Architectures
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Executive Summary

This workflow demonstrates how Amazon Bedrock can be integrated with Gmail and Google Sheets to build an AI-powered customer support solution. Incoming emails are automatically analyzed, company policies are retrieved from a knowledge base, responses are generated using Claude, and customers receive timely, professional support responses.